

Lecture 26. Reduction to Hessenberg or Tridiagonal Form

We now describe the first of the two computational phases outlined in the previous lecture: reduction of a full matrix to Hessenberg form by a sequence of unitary similarity transformations. If the original matrix is hermitian, the result is tridiagonal.

A Bad Idea

To compute the Schur factorization $A = Q T Q^*$, we would like to apply unitary similarity transformations to A in such a way as to introduce zeros below the diagonal. A natural first idea might be to attempt direct triangularization by using Householder reflectors to introduce these zeros, one after another.

The first Householder reflector Q_1^* , multiplied on the left of A , would introduce zeros below the diagonal in the first column of A . In the process it will change all of the rows of A . In this and the following diagrams, as usual, entries that are changed at each step are written in boldface:

$$\begin{array}{ccc} \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix} & \xrightarrow{Q_1^*} & \begin{bmatrix} \mathbf{\times} & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} \\ 0 & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} \\ 0 & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} \\ 0 & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} \\ 0 & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} & \mathbf{\times} \end{bmatrix} \\ A & & Q_1^* A \end{array}$$

Unfortunately, to complete the similarity transformation, we must also multiply by Q_1 on the right of A :

$$\begin{array}{ccc} \begin{bmatrix} \times & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & & \times & \times & \times \\ & & & \times & \times \\ & & & & \times \end{bmatrix} & \xrightarrow{\cdot Q_1} & \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix} \\ Q_1^* A & & Q_1^* A Q_1 \end{array}$$

This has the effect of replacing each column of the matrix by a linear combination of all the columns. The result is that the zeros that were previously introduced are destroyed; we are no better off than when we started.

Of course, with hindsight we know that this idea had to fail, because of the “fundamental difficulty” described in the previous lecture. No finite process can reveal the eigenvalues of A exactly.

Curiously, this too-simple strategy, which appears futile as we have discussed it, does have the effect, typically, of reducing the size of the entries below the diagonal, even if it does not make them zero. We shall return to this “bad idea” when we discuss the QR algorithm.

A Good Idea

The right strategy for introducing zeros in Phase 1 is to be less ambitious and operate on fewer entries of the matrix. We shall only conquer territory we are sure we can defend.

At the first step, we select a Householder reflector Q_1^* that leaves the first row unchanged. When it is multiplied on the left of A , it forms linear combinations of only rows $2, \dots, m$ to introduce zeros into rows $3, \dots, m$ of the first column. Then, when Q_1 is multiplied on the right of $Q_1^* A$, it leaves the first column unchanged. It forms linear combinations of columns $2, \dots, m$ and does not alter the zeros that have been introduced:

$$\begin{array}{ccc} \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix} & \xrightarrow{Q_1^*} & \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times \end{bmatrix} \\ A & & Q_1^* A \end{array} \quad \xrightarrow{\cdot Q_1} \quad \begin{array}{ccc} \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix} & & \\ Q_1^* A Q_1 & & \end{array}$$

This idea is repeated to introduce zeros into subsequent columns. For example, the second Householder reflector, Q_2 , leaves the first and second rows and

columns unchanged:

$$\begin{array}{ccc}
 \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & & & & \times \end{bmatrix} & \xrightarrow{Q_2^*} & \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & 0 & \times & \times & \times \\ & 0 & \times & \times & \times \end{bmatrix} & \xrightarrow{\cdot Q_2} & \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & \times & \times & \times & \times \\ & & & & \times \end{bmatrix} \\
 Q_1^* A Q_1 & & Q_2^* Q_1^* A Q_1 & & Q_2^* Q_1^* A Q_1 Q_2
 \end{array}$$

After repeating this process $m - 2$ times, we have a product in Hessenberg form, as desired:

$$\underbrace{Q_{m-2}^* \cdots Q_2^* Q_1^*}_{Q^*} A \underbrace{Q_1 Q_2 \cdots Q_{m-2}}_Q = H.$$

The algorithm is formulated below; compare Algorithm 10.1.

Algorithm 26.1. Householder Reduction to Hessenberg Form

for $k = 1$ **to** $m - 2$

$$x = A_{k+1:m,k}$$

$$v_k = \text{sign}(x_1) \|x\|_2 e_1 + x$$

$$v_k = v_k / \|v_k\|_2$$

$$A_{k+1:m,k:m} = A_{k+1:m,k:m} - 2v_k(v_k^* A_{k+1:m,k:m})$$

$$A_{1:m,k+1:m} = A_{1:m,k+1:m} - 2(A_{1:m,k+1:m} v_k) v_k^*$$

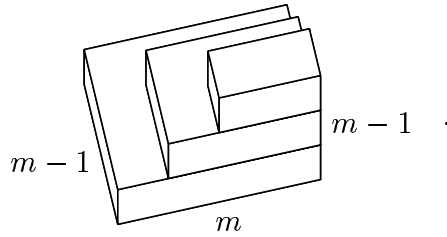
Just as in Algorithm 10.1, here the matrix $Q = \prod_{k=1}^{m-2} Q_k$ is never formed explicitly. The reflection vectors v_k are saved instead, and can be used to multiply by Q or reconstruct Q later if necessary. For details, see Lecture 10.

Operation Count

The number of operations required by Algorithm 26.1 can be counted with the same geometric reasoning we have used before. The rule of thumb is that unitary operations require four flops for each element operated upon.

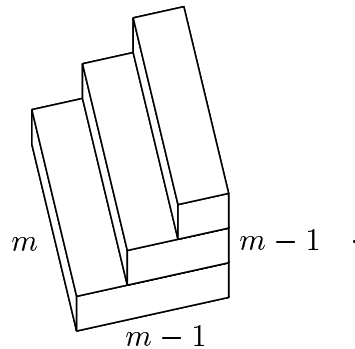
The work is dominated by the two updates of submatrices of A . The first loop applies a Householder reflector on the left of the matrix. The k th such reflector operates on the last $m - k$ rows. Since at the time the reflector is applied, these rows have zeros in the first $k - 1$ columns, arithmetic has to be

performed only on the last $m - k + 1$ entries of each row. The picture is as follows:



As $m \rightarrow \infty$, the volume converges to $\frac{1}{3}m^3$. At four flops per element, the amount of work in this loop is $\sim \frac{4}{3}m^3$ flops.

The second inner loop applies a Householder reflector on the right of the matrix. At the k th step, the reflector operates by forming linear combinations of the last $m - k$ columns. This loop involves more work than the first one because there are no zeros that can be ignored. Arithmetic must be performed on all of the m entries of each of the columns operated upon, a total of $m(m-k)$ entries for a single value of k . The picture looks like this:



The volume converges as $m \rightarrow \infty$ to $\frac{1}{2}m^3$, so, at four flops per element, this second loop requires $\sim 2m^3$ flops.

All together, the total amount of work for unitary reduction of an $m \times m$ matrix to Hessenberg form is:

$$\text{Work for Hessenberg reduction: } \sim \frac{10}{3}m^3 \text{ flops.} \quad (26.1)$$

The Hermitian Case: Reduction to Tridiagonal Form

If A is hermitian, the algorithm just described will reduce A to tridiagonal form (at least, in the absence of rounding errors). This is easy to see: since A is hermitian, Q^*AQ is also hermitian, and any hermitian Hessenberg matrix is tridiagonal.

Since zeros are now introduced in rows as well as columns, additional arithmetic can be avoided by ignoring these additional zeros. With this optimization, applying a Householder reflector on the right is as cheap as applying

the reflector on the left, and the total cost of applying the right reflectors is reduced from $2m^3$ to $\frac{4}{3}m^3$ flops. We have two pyramids to add up instead of a pyramid and a prism, and the total amount of arithmetic is reduced to $\frac{8}{3}m^3$ flops.

This saving, however, is based only on sparsity, not symmetry. In fact, at every stage of the computation, the matrix being operated upon is hermitian. This gives another factor of two that can be taken advantage of, bringing the total work estimate to

$$\text{Work for tridiagonal reduction: } \sim \frac{4}{3}m^3 \text{ flops.} \quad (26.2)$$

We shall not give the details of the implementation.

Stability

Like the Householder algorithm for QR factorization, the algorithm just described is backward stable. Recall from Theorem 16.1 that, for any $A \in \mathbb{C}^{m \times n}$, the Householder algorithm for QR factorization computes reflection vectors equivalent to an implicit, exactly unitary factor $\tilde{Q}$ (16.2), as well as an explicit upper-triangular factor $\tilde{R}$, such that

$$\tilde{Q}\tilde{R} = A + \delta A, \quad \frac{\|\delta A\|}{\|A\|} = O(\epsilon_{\text{machine}}).$$

The same kind of error estimate can be established for Algorithm 26.1. Let $\tilde{H}$ be the actual Hessenberg matrix computed in floating point arithmetic, and let $\tilde{Q}$, as before, be the exactly unitary matrix (16.2) corresponding to the reflection vectors $\tilde{v}_k$ computed in floating point arithmetic. The following result can be proved.

Theorem 26.1. *Let the Hessenberg reduction $A = QHQ^*$ of a matrix $A \in \mathbb{C}^{m \times m}$ be computed by Algorithm 26.1 on a computer satisfying the axioms (13.5) and (13.7), and let the computed factors $\tilde{Q}$ and $\tilde{H}$ be defined as indicated above. Then we have*

$$\tilde{Q}\tilde{H}\tilde{Q}^* = A + \delta A, \quad \frac{\|\delta A\|}{\|A\|} = O(\epsilon_{\text{machine}}) \quad (26.3)$$

for some $\delta A \in \mathbb{C}^{m \times m}$.

Exercises

26.1. Theorem 26.1 and its successors in later lectures show that we can compute eigenvalues $\{\tilde{\lambda}_k\}$ of A numerically that are the exact eigenvalues of a

matrix $A + \delta A$ with $\|\delta A\|/\|A\| = O(\epsilon_{\text{machine}})$. Does this mean they are close to the exact eigenvalues $\{\lambda_k\}$ of A ? This is a question of eigenvalue perturbation theory.

One can approach such problems geometrically as follows. Given $A \in \mathbb{C}^{m \times m}$ with spectrum $\Lambda(A) \subseteq \mathbb{C}$ and $\epsilon > 0$, define the 2-norm ϵ -pseudospectrum of A , $\Lambda_\epsilon(A)$, to be the set of numbers $z \in \mathbb{C}$ satisfying any of the following conditions:

- (i) z is an eigenvalue of $A + \delta A$ for some δA with $\|\delta A\|_2 \leq \epsilon$;
- (ii) There exists a vector $u \in \mathbb{C}^m$ with $\|(A - zI)u\|_2 \leq \epsilon$ and $\|u\|_2 = 1$;
- (iii) $\sigma_m(zI - A) \leq \epsilon$;
- (iv) $\|(zI - A)^{-1}\|_2 \geq \epsilon^{-1}$.

The matrix $(zI - A)^{-1}$ in (iv) is known as the *resolvent* of A at z ; if z is an eigenvalue of A , we use the convention $\|(zI - A)^{-1}\|_2 = \infty$. In (iii), σ_m denotes the smallest singular value.

Prove that conditions (i)–(iv) are equivalent.

26.2. Let A be the 32×32 matrix with -1 on the main diagonal, 1 on the first and second superdiagonals, and 0 elsewhere.

(a) Using an SVD algorithm built into MATLAB or another software system, together with contour plotting software, generate a plot of the boundaries of the 2-norm ϵ -pseudospectra of A for $\epsilon = 10^{-1}, 10^{-2}, \dots, 10^{-8}$.

(b) Produce a semilogy plot of $\|e^{tA}\|_2$ against t for $0 \leq t \leq 50$. What is the initial growth rate of the curve before the eventual decay sets in? Can you relate this to your plot of pseudospectra? (Compare Exercise 24.3.)

26.3. One of the best known results of eigenvalue perturbation theory is the *Bauer–Fike theorem*. Suppose $A \in \mathbb{C}^{m \times m}$ is diagonalizable with $A = V\Lambda V^{-1}$, and let $\delta A \in \mathbb{C}^{m \times m}$ be arbitrary. Then every eigenvalue of $A + \delta A$ lies in at least one of the m circular disks in the complex plane of radius $\kappa(V)\|\delta A\|_2$ centered at the eigenvalues of A , where κ is the 2-norm condition number. (Compare Exercise 24.2.)

(a) Prove the Bauer–Fike theorem by using the equivalence of conditions (i) and (iv) of Exercise 26.1.

(b) Suppose A is normal. Show that for each eigenvalue $\tilde{\lambda}_j$ of $A + \delta A$, there is an eigenvalue λ_j of A such that

$$|\tilde{\lambda}_j - \lambda_j| \leq \|\delta A\|_2. \quad (26.4)$$